INCH-POUND
MIL-PRF-3098/61K
7 January 2010
SUPERSEDING
MIL-PRF-3098/61J
12 January 2005

PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR84/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-3098.

Pertinent characteristics: 17 MHz to 61 MHz; third mechanical overtone; noncontrolled; series resonance.

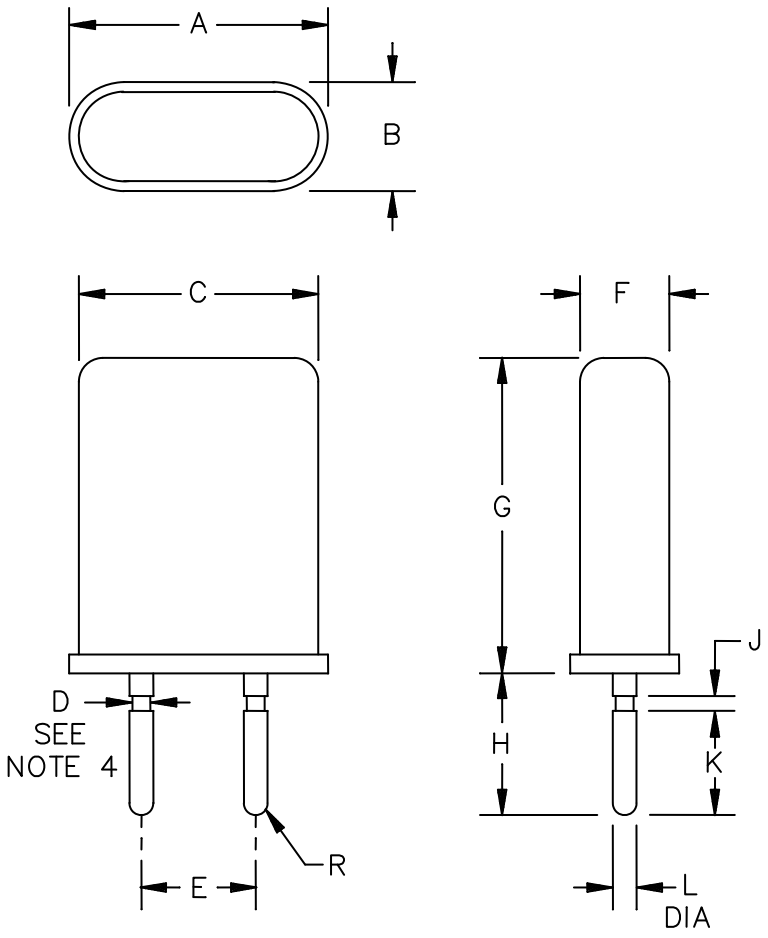

FIGURE 1. Crystal unit - CR84/U.

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Ltr	Inches		mm	
	Min	Max	Min	Max
A	---	.435	---	11.05
B	---	.183	---	4.65
C	---	.402	---	10.21
D	.027	.032	0.69	0.81
E	.184	.200	4.67	5.08
F	---	.150	---	3.81
G	---	.530	---	13.46
H	.223	.248	5.66	6.30
J	.015	.025	0.38	0.64
K	.165	.175	4.19	4.45
L	.038	.042	0.97	1.07

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-PRF-3098.
4. Pin undercut may be omitted.

FIGURE 1. Crystal unit - CR84/U - Continued.

REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 17.0 MHz to 61.0 MHz, inclusive.

Frequency tolerances:

Primary operating temperature range: ±20 parts per million (ppm).

Room temperature range: ±70 ppm.

Frequency stability: ±5 ppm

Equivalent resistance: 40 ohms, maximum.

Drive level: 1.0 mW maximum.

Resonance: Series.

Mode of oscillation: Third mechanical overtone.

Reference temperature: -85°C ±1°C.

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Operating temperature ranges:

Operable: -55°C to +80°C, and +90°C to +105°C, inclusive.

Operating: +80°C to +90°C, inclusive.

Capacitance, shunt: 7.0 pF, maximum.

Shock (specified pulse):

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Vibration: Method 204 of MIL-STD-202, test condition A.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Aging:

Frequency change permitted: ±5 ppm.

Part or Identifying Number (PIN): CR84/U (followed by specified frequency), see 1.2 of MIL-PRF-3098.

Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202

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Army - CR
Navy - EC
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DLA - CC

Review activities:
Army - AR, MI
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Preparing activity:
Army - CR

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